



NEWSGENIE

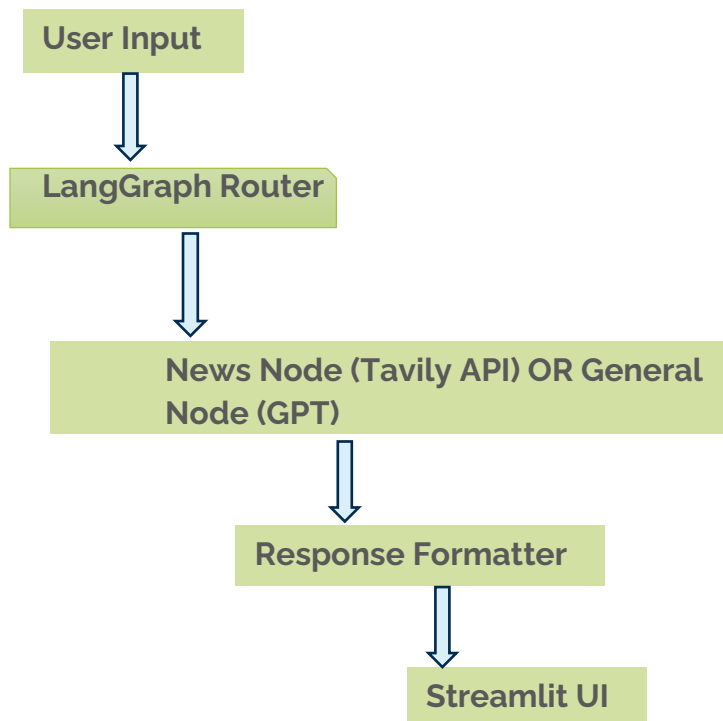
AN AI-POWERED INFORMATION AND NEWS
ASSISTANT

1. SOLUTION OVERVIEW –

NewsGenie integrates multiple AI components:

- OpenAI GPT – conversational responses
- Tavily API – real-time news retrieval
- LangGraph – workflow orchestration
- Streamlit – interactive UI
- Intelligent routing between news & general queries

2. System Architecture -



3. LangGraph-Based Workflow

Router node analyzes user input

- If keywords like 'news' or 'latest' detected → News Node
- Otherwise → General Node
- Each node returns structured state
- Workflow ends with compiled agent execution
- Ensures modular and scalable query handling

4. API Integration

Tavily API:

- Real-time news retrieval
- Advanced search depth
- Structured results & caching

OpenAI GPT:

- Conversational responses
- Secure environment variable integration

5. Error Handling Strategies

Environment variable validation

- Try-except blocks for API failures
- Graceful fallback messages
- No-results handling
- Controlled system failure responses

6. User Interface Design

Streamlit-based interface

- Text input for user queries
- Real-time response rendering
- Session history tracking
- Loading spinner & clear error messaging

7. Fallback Mechanisms

- Ensuring System Reliability During Failures
 - ❖ Environment variable validation at startup
 - ❖ Prevents execution without required API keys
 - ❖ Try-except handling for external API failures
 - ❖ Graceful fallback messages for users
 - ❖ No-results handling to avoid blank responses
 - ❖ Controlled error display (no technical stack traces)
- Environment & API Validation

- ❖ Validates OPENAI_API_KEY and TAVILY_API_KEY
- ❖ Stops application safely if keys are missing
- ❖ Prevents runtime authentication failures
- ❖ Ensures secure API integration
- Tavily API Failure Handling
 - ❖ Handles network errors and downtime
 - ❖ Catches rate limit exceptions
 - ❖ Returns: "Service temporarily unavailable"
 - ❖ Maintains UI responsiveness
 - ❖ Prevents system crash
- No-Results Handling
 - ❖ Detects empty API responses
 - ❖ Returns: "No relevant news found"
 - ❖ Prevents blank screens
 - ❖ Maintains conversational flow